PROJECT GUIDE v5 · AASAN ROMAN URDU

SENTINEL

Fire & Emergency Monitoring System

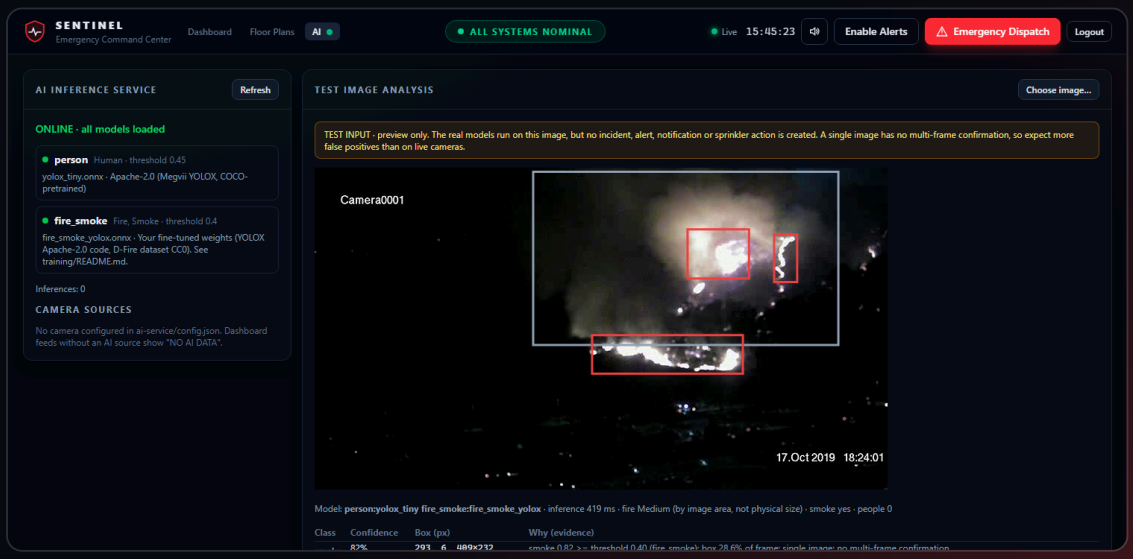
Poore project ka flow shuru se aakhir tak: camera, AI, alerts, sensors, sprinklers, naqsha aur evacuation. AI models ki har detail: model kahan se aaya, dataset kaun sa aur kahan se, kitni tasveerein, kitne epochs, kahan train hua, aur asli accuracy.

AI Models
YOLOX-Tiny · D-Fire 21,527 tasveerein · 80 epochs (Kaggle T4)

Accuracy
Aag 93% · Dhuwan 92% pakda · galat alarm 0.6%

System
Angular · ASP.NET Core 8 · SQL Server · Python AI

Status
✅ 22 · ⚠️ 4 · ❌ 0 (26 requirements)



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1. Sentinel Kya Hai?

Ek line mein: Sentinel ek **fire-safety control room software** hai. Ye building ke cameras (CCTV aur drone), dhuwen/garmi ke sensors aur smart sprinklers ko ek screen par jorta hai. **AI** camera mein aag, dhuwan aur insaan pehchaanta hai. Khatra ho to system khud:

- screen par **alarm** aur alert dikhata hai,
- **SMS, Email aur browser Push** notification bhejta hai,
- us zone ke **sprinklers** chala sakta hai,
- har waqia ka **log** rakhta hai,
- building ke **naqshe par mehfooz raasta** aur bahar nikalne ka andazan waqt batata hai.

Asal zindagi ki misaal

Ek factory ka security room sochiye: deewar par TV screens, aur ek guard jo sab dekh raha hai. Sentinel us guard ka computer version hai:

Asal duniya	Sentinel mein
CCTV ki screen	Dashboard ki **Video Wall** (Angular)
Guard ki aankh jo aag pehchaanti hai	**AI Service** (Python, YOLOX models)
Guard ka dimagh: "kitna khatarnak hai?"	`AlertEngine.cs` (backend)
Guard ki notebook	SQL Server ke `IncidentAlerts` aur `IncidentLogs` tables
Guard ka phone	`NotificationWorker`: Twilio SMS, Email, Web Push
Sprinkler ka switch	`SprinklerService.cs` + `SmartSprinklers` table
Radio jis se sab ko foran khabar milti hai	**SignalR** (live connection)
Building ka naqsha deewar par	**Floor Plans** editor + Evacuation route

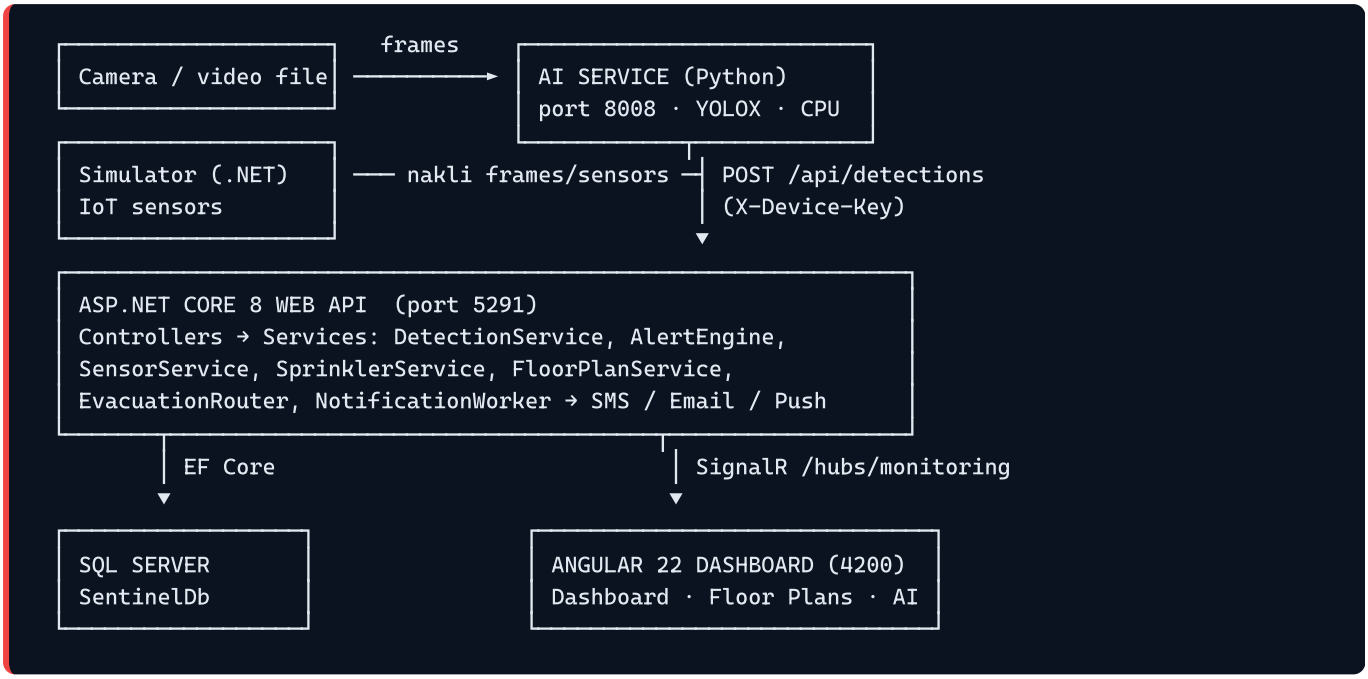
Zaroori alfaaz (aasan matlab)

Lafz	Matlab
API	Backend ka "darwaza" jahan request aati hai, jaise `POST /api/detections`
SignalR	Hamesha khula live connection. Server browser ko khud data bhej deta hai, page refresh ki zaroorat nahi
JWT token	Login ke baad milne wala "digital pass", har request ke saath jata hai
Device Key	Machines (AI service, sensors, simulator) ka password, header `X-Device-Key`
Incident	Ek khatre ka waqia (aag, dhuwan, khatre mein insaan, sensor alarm)
Severity	Khatre ka darja: Low < Medium < High < Critical
AI model	Training se seekha hua program jo tasveer mein cheezein pehchaanta hai
Dataset	Labels wali tasveeron ka majmua jis se model seekhta hai

Lafz	Matlab
Epoch	Training mein poore dataset par **ek chakkar**
ONNX	AI model ki file ka format jo bina PyTorch ke CPU par chalta hai
Simulated	Asli nahi, test data. Aisi har cheez par **SIMULATED** label lagta hai

⚠️ **Imandari ki baat:** sab kuch software aur lab mein test hua hai. Asli CCTV camera, asli SMS/Email aur asli sprinkler hardware ke saath test **nahi** hua. Is liye ye system abhi asli emergency ke liye **"production-safe" nahi** kaha ja sakta. Smoke detectors, fire panel aur insaani nigrani zaroori hain.

2. Poora System Ek Nazar Mein



Hissa	Technology	Folder	Port
Frontend	Angular 22, TypeScript, Tailwind CSS 4, Canvas, hls.js	`frontend/`	4200
Backend	ASP.NET Core 8 Web API (C#), SignalR, EF Core 8	`backend/Sentinel.Api/`	5291
Database	Microsoft SQL Server, EF migrations	`backend/Sentinel.Api/Migrations/`	1433
AI Service	Python 3.13, ONNX Runtime (CPU), OpenCV, NumPy	`ai-service/`	8008 (sirf 127.0.0.1)
Simulator	.NET console: nakli sensors, sprinklers aur AI frames	`backend/Sentinel.Simulator/`	—
Notifications	Twilio SMS, SMTP Email (MailKit), Web Push (VAPID)	`backend/Sentinel.Api/Services/Notifications/`	—
Tests	xUnit, Vitest, Python unittest	`backend/Sentinel.Api.Tests/`, `frontend/src/**/*.spec.ts`, `ai-service/tests/`	—

Tabaqay (layers): Angular Component → Angular Service → HTTP (JWT) → Controller → Service → EF Core → SQL Server.

Waiter aur chef ki misaal: Controller "waiter" hai jo order leta hai. Service "chef" hai jo asal kaam karti hai. EF Core "translator" hai jo C# ko SQL mein badalta hai.

3. Poora Flow: Camera Se Screen Tak

Misaal: warehouse ke camera mein aag lagi

```
1. Camera frame          → AI Service (sentinel_ai) har second 1–4 frame leta hai
2. AI Service            → person + fire/smoke model chalata hai, boxes nikalta hai
3. Tracking              → har cheez ko track id (F11, S2, P3)
4. Confirmation          → wohi aag 3 frames mein dikhi? tab hi report
5. Severity              → aag ke boxes frame ka kitna % → Low/Medium/High/Critical
6. AI danger zone        → aag ke gird 25% bara laal zone
7. POST /api/detections  → API (X-Device-Key ke saath)
8. DetectionPayloadValidator → JSON ki jaanch + insaan danger zone mein? (backend khud hisaab)
9. RealtimeNotifier      → SignalR "detection" → sab dashboards par boxes
10. AlertEngine          → incident banao/barhao (dedupe ke saath)
11. IncidentLogs         → Opened, FireMeasurement, PersonInDanger
12. Emergency?           → SMS/Email/Push queue + Auto sprinklers (simulated data par nahi)
13. SignalR              → "incident", "incidentLog", "activity" → dashboard update
14. Evacuation panel     → us zone ka danger zone active → naya mehfooz raasta
15. Calibrated camera    → camera mein dikha insaan naqshe par amber nuqta (Chapter 9)
16. Risk prediction      → aag kin kamron tak pahunch sakti hai + naqshe par risk heatmap (Chapter
9)
```

Bina refresh ke kaise? Browser aur server ke darmiyan SignalR connection hamesha khula rehta hai. Server naya data milte hi sab browsers ko bhej deta hai. Canvas har animation frame (~60 dafa per second) par dobara draw hota hai.

Detection do raaston se aa sakti hai:

- REST: `POST /api/detections` (AI service yehi use karti hai).
- SignalR: hub method `PublishDetection(payload)` (simulator). Hub par 60 messages per second ki had hai.

4. Alert Engine: Incident Kaise Banta Hai

File: `backend/Sentinel.Api/Services/AlertEngine.cs`

Severity (khatre ka darja)

Halat	Severity
Sirf dhuwan (aag nahi)	Medium
Aag	AI ki `fireSeverity` (na ho to Medium)
Aag + koi insaan nazar aaye	Kam az kam High
1–2 insaan danger zone mein	High
3 ya zyada insaan danger zone mein	Critical
Aag + koi insaan danger zone mein	**Critical**

Emergency ka rule

```
Emergency = DangerCount > 0   YA   aag ki severity Critical   YA   incident Critical
```

Emergency par do kaam **khud** hote hain:

1. **SMS + Email + Push** notification queue mein.
2. Us zone ke **Auto mode** sprinklers ON. Lekin agar data **SIMULATED** hai to sprinkler **nahi** chalte (`Alerts:AutoSprinklerOnSimulatedData = false`), sirf log likha jata hai.

Duplicate se bachao

Tareeqa	Kya karta hai	Setting
Merge window	Same source ka khula incident 90 second mein dobara aaye to naya nahi, wohi update	`IncidentMergeWindowSeconds = 90`
Escalation	Severity sirf barhti hai, kam nahi hoti	—
Vision throttle	Ek camera ka database check har 2 second mein ek dafa (severity barhe to foran)	`VisionEvaluationIntervalSeconds = 2`
Cooldown	Same source + zone + severity par 120 second tak dobara SMS nahi	`CooldownSeconds = 120`
Lock	Do frames ek saath aayein to bhi ek hi incident	`IncidentLocks`

Camera lost (camera chup ho jaye)

Aag mein bijli jaye ya cable kate to camera frames bhejna band kar deta hai. `CameraWatchdog` har 5 second mein dekhta hai:

Halat	Kya hota hai
Jo camera frames bhej raha tha, **30 second** se chup	"Camera lost" incident (source **Camera lost**): **Medium**
Usi waqt us zone mein aag ka incident khula	**High**, message mein "while a fire incident is open in this zone"
Camera dobara frames bhejne lage	Live Event Log: "Camera … is sending frames again"
Camera kabhi juda hi nahi (API chalne ke baad)	Koi alarm nahi

- Har chup rehne par **ek** alert, aur dobara chup ho to wohi khula incident update hota hai.
- Setting: `Alerts:CameraLostSeconds = 30`. Code: `Services/CameraWatchdog.cs`, `AlertEngine.CameraLostAsync`.
- Ye **High** hai, emergency nahi, is liye SMS khud nahi jata. Operator dashboard par dekhta hai.
- Had: last-frame ka waqt API ki memory mein rehta hai. API restart ke baad camera ka pehla frame aane tak use watch nahi kiya jata.

5. Sensors Aur Smart Sprinklers

Sensors (dhuwan ppm, garmi °C)

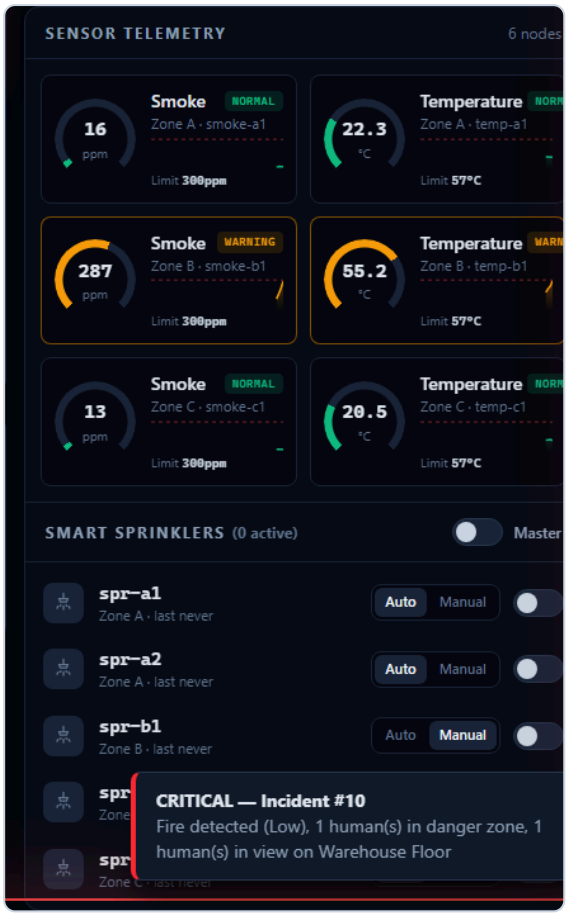
File: `Services/SensorRules.cs`

Halat	Shart (misaal: temperature threshold 57°C)	Kya hota hai
Normal	45.6°C se kam	Kuch nahi
Warning	≥ threshold × 0.8	Peela rang
Alarm	≥ threshold	High incident + zone ke Auto sprinklers
Critical	≥ threshold × 1.25 (smoke × 1.5)	Critical incident + **emergency notifications** + sprinklers

- Readings `SensorReadings` table mein, **48 ghante** se purani khud delete.
- **Test data:** telemetry batch mein `"simulated": true` ho (simulator yehi bhejta hai) to incident par **[SIMULATED DATA]** likha aata hai aur sprinklers **khud nahi** chalte (`Alerts:AutoSprinklerOnSimulatedData = false`).
- Dashboard par gauge aur chhota graph (sparkline), aakhri 60 readings.

Smart sprinklers

- Har sprinkler ka ek **zone** hai (Zone A/B/C).
- **Auto mode:** Alert engine khud ON kar sakta hai.
- **Manual mode:** sirf operator ON/OFF karta hai.
- **Master switch:** saare sprinklers ek click mein, pehle confirm dialog.
- Sprinkler khud OFF **nahi** hota. Aag bujhne ke baad operator OFF karta hai.



⚠️ Is project mein sprinklers **simulated** hain (database ki entries). Koi asli hardware nahi joda gaya.

6. Notifications (SMS, Email, Push)

```
AlertEngine ──Enqueue──► NotificationDispatcher (queue + cooldown, foran wapas)
                                  ▼
                        NotificationWorker (background thread)
                                  ├─► TwilioSmsChannel → SMS
                                  ├─► EmailChannel     → Email (MailKit / SMTP)
                                  └─► WebPushChannel   → Browser push
                                  ▼
                        SignalR "notification" → Live Event Log mein natija
```

- **Background mein kyun?** SMS/Email mein 2–5 second lag sakte hain. Request ke andar hota to live video atak jati.
- **Kab jati hain?** Sirf **emergency** par, ya jab operator **Emergency Dispatch** dabaye.
- **Setting na ho to?** Channel "skipped (not configured)" likhta hai, crash nahi.
- **Email console mode:** SMTP host khali ho aur recipients hon to email API ki terminal mein print hota hai.
- Credentials `.env` mein (git se bahar). `.env.example` mein sirf namuna.

> ⚠️ Is computer ki `.env` mein **asli** Twilio/SMTP credentials hain, lekin asli SMS/Email bhej kar **test nahi** kiya gaya.

7. Danger Zone, Safe Zone Aur Insaan Ki Hifazat

Danger zone person check (backend khud karta hai)

File: `Validators/DetectionPayloadValidator.cs`

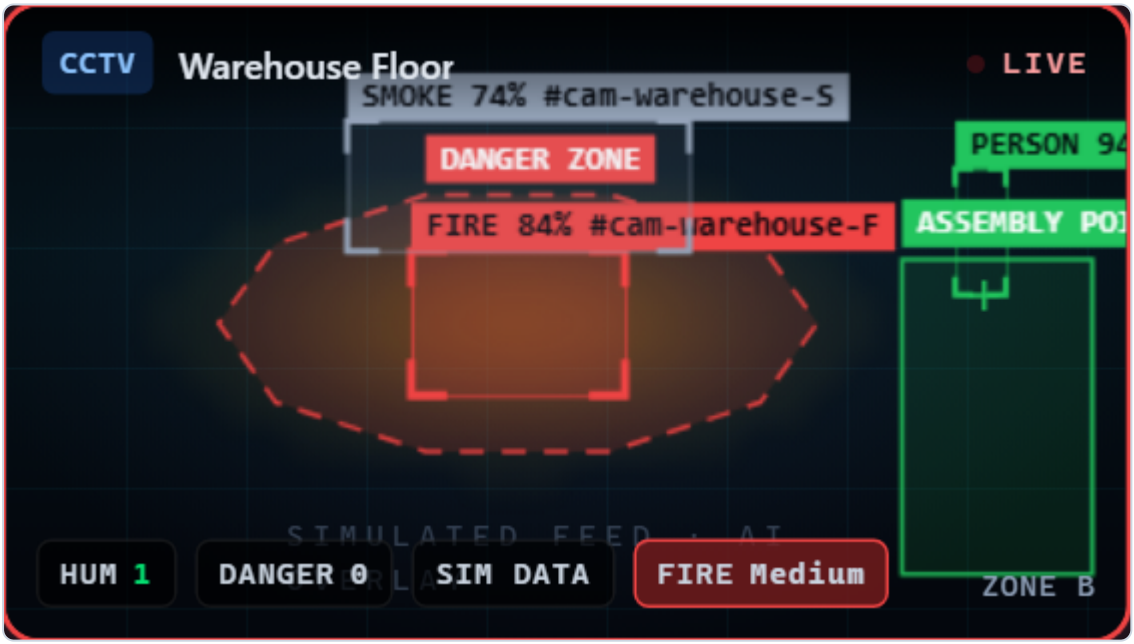
1. Har insaan ke box ka **pair ka point**: `x + width/2, y + height` (box ka neeche beech).
2. Ye point kisi **Danger** ya **Critical** zone polygon ke andar hai? (ray-casting). Warning aur Safe zones nahi gine jaate.
3. Haan → `AtRisk = true`, zone ka naam, aur `DangerCount` barhta hai. Backend kabhi AI se kam nahi ginta.
4. Har (incident, insaan, zone) ka sirf **ek** `PersonInDanger` log.

Danger zones kahan se aate hain

- **AI:** har confirmed aag ke box ko 25% bara karke "AI fire area" zone.
- **Camera config:** `ai-service/config.json` mein har camera ke fixed zones (misaal "Fuel store").
- **Naqsha:** Floor plan ke danger zones (Chapter 9).

Safe zones (hare rang)

Mehfooz ilaqe (assembly point) **hare**, solid line. Danger zones **laal**, dashed aur dhadakte hue.



Boxes ke rang

Cheez	Rang
Aag	Laal, dhadakta
Dhuwan	Gray
Insaan (mehfooz)	Hara

Cheez	Rang
Insaan danger zone mein	Amber, "IN DANGER"
Heatmap	Peela → laal

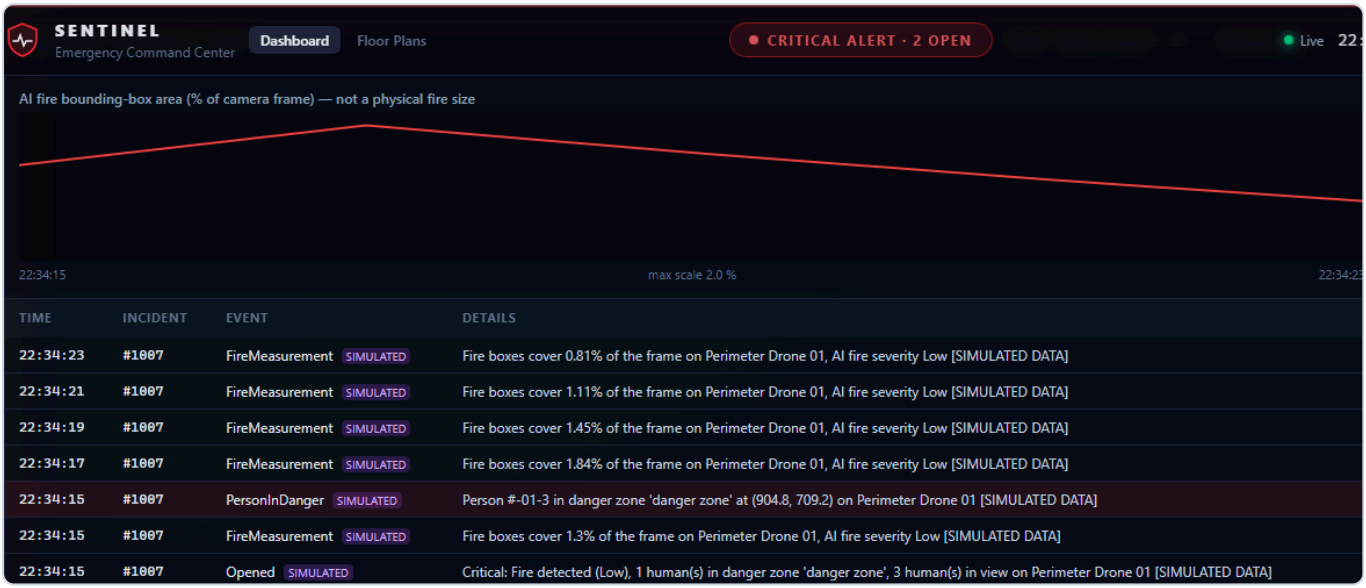
AI 1.5 second tak data na bheje to tile par **"NO AI DATA"**. 4.5 second baad boxes gayab.

8. Fire Growth History Aur Logs

Aag ki naap (graph)

```
FireAreaPercent = (saare Fire boxes ka width × height ka jama) ÷ (frame ki width × height) × 100
```

- Camera incident: AI fire area % ka graph. Sensor incident: sensor readings ka graph.
- Graph par likha hai **"not a physical fire size"**: ye screen par hissa hai, meters nahi.



Logs (`IncidentLogs` table)

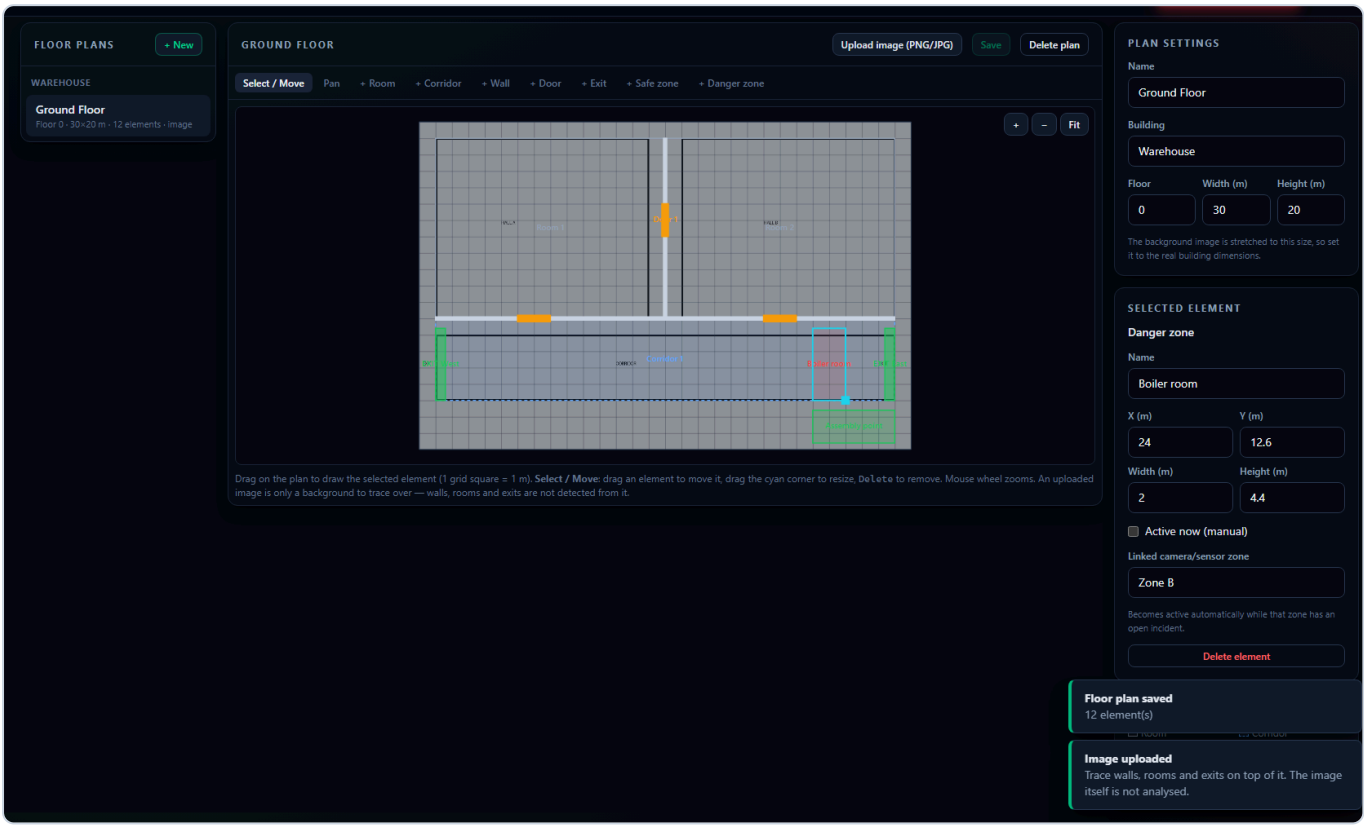
Type	Kab
`Opened`	Naya incident
`Escalated`	Darja barha
`Resolved`	Operator ne Resolve dabaya
`FireMeasurement`	Aag ki naap badli
`PersonInDanger`	Insaan danger zone mein
`EvacuationRoute`	Raasta nikala gaya ("ESTIMATE")

Har log mein `Simulated` column. Dashboard par SIMULATED (jamni) aur ESTIMATE (neela) labels. Naya log SignalR `incidentLog` se foran screen par.


ALERT LOGS
Open (4)
All
TIME SEVERITY LOCATION SOURCE FIRE HUMANS IN DANGER STATUS
13 21:02:24 HIGH Zone B (sensor temp-b1) Sensor – 0 0 Open Resolve
12 21:02:24 HIGH Zone B (sensor smoke-b1) Sensor – 0 0 Open Resolve
11 21:02:11 CRITICAL Zone B (Perimeter Drone 01) AI Vision High 3 1 Open Resolve
10 21:02:11 CRITICAL Zone B (Warehouse Floor) AI Vision High 1 1 Open Resolve

9. Floor Plans, Evacuation Raasta Aur Waqt

Naqsha banana (Floor Plans page)



Tool	Kaam
+ Room / + Corridor	Chalne ki jagah
+ Wall	Deewar (0.2 m moti)
+ Door	Darwaza (deewar par). "Blocked" = band
+ Exit	Emergency exit
+ Safe zone / + Danger zone	Mehfooz / khatarnak ilaqa. Danger zone ko camera zone (jaise `Zone B`) se jor sakte hain

- Apne naqshe ki **PNG/JPG** background mein upload kar sakte hain (5 MB tak, file ke andar ke bytes check hote hain).
- Tasveer sirf background hai. System tasveer se deewarein **khud nahi** pehchaanta.
- Har cheez meters mein, 0.1 m par snap. Backend 2,000 elements tak, ghalti par 400 aur saaf message.

Raasta kaise nikalta hai (`EvacuationRouter.cs`)

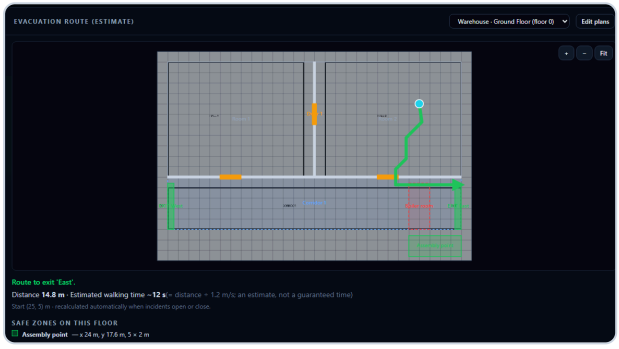
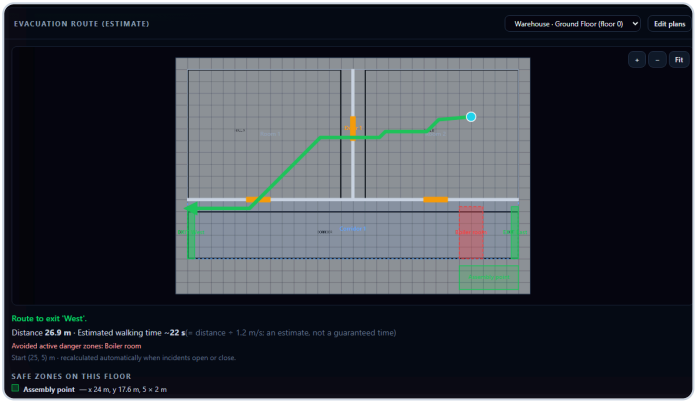
1. Naqsha 0.5 m ke khanon (grid) mein. Bari building par khana bara (max ~200,000 khane).
2. Chal sakte hain: khule Room/Corridor/Door/Exit/SafeZone. Band: deewar, band darwaza, **active danger zone**.
3. Danger zone active hai jab "Active now" tick ho, **ya** us ke linked zone mein khula incident ho.

4. **Dijkstra** (8 simtein): qareeb tareen pohanch-sakne-wala khula exit.
5. **1 m clearance:** danger zone se 1 m ke andar wale khane mehngay. Koi raasta 1 m door na reh sake to saaf warning: "CLEARANCE VIOLATED".
6. Insaan khud danger zone ke andar ho to bahar nikalne ki ijazat, message "leave immediately".

Waqt (ETA)

```
Andazan waqt = faasla (m) ÷ 1.2 m/s       misaal: 26.9 m ÷ 1.2 = ~22 s
```

Dhuwan, bheer, seedhiyan shamil nahi. Is liye screen par "estimate" likha hai.

Aag se pehle	Zone B mein aag ke baad
	

Aag kahan phailegi (risk prediction + risk heatmap)

Requirement (Part 3): "Predict which areas might be affected next" aur "Generate heatmaps for risk zones", taake high-risk ilaqon ki **early warning** mil sake.

Kaise hisaab hota hai (rule-based, `Services/FireRiskEstimator.cs`):

1. Aag = naqshe ke **active danger zones**: "Active now" tick, ya linked zone (jaise `Zone B`) mein khula incident.
2. Naqsha 1 m ke khanon mein. Aag har khane tak **kamron, corridors aur darwazon** se kitni door hai, ye nikalta hai. **Deewar rokti hai.** Band darwaza nahi rokta, kyunki band darwaza aag ko sirf dheema karta hai.
3. Faasle se darja:

Darja	Faasla aag se	Rang
Burning	Danger zone ke andar	Gehra laal
High	≤ 5 m	Laal
Medium	≤ 10 m	Narangi
Low	≤ 20 m	Peela
None	20 m se zyada, ya deewar ke peeche (darwaza nahi)	Koi rang nahi

4. Har kamre aur corridor ka darja uske sab se qareeb hisse se, aur **"Next at risk"** mein sab se qareeb High/Medium wale 3 kamre.



Dashboard par: Evacuation panel mein naqshe ke upar heatmap, aur neeche "⚠ Fire risk: Next at risk: Main corridor (7.4 m, Medium), Hall B (8 m, Medium)". Kisi zone mein incident khule ya band ho to ye **khud** dobara hisaab hota hai.

Cheez	Tafseel
API	`GET /api/floorplans/{id}/risk`
Jawab	Har kamre ka darja aur faasla, message, aur heatmap ka grid (`cells` : -1 bahar, 0 none, 1 low, 2 medium, 3 high, 4 burning)
Code	`FireRiskEstimator.cs` (backend), `evacuation-panel` (frontend)

> ⚠️ **Imandari ki baat:** ye **rule-based andaza** hai, AI model (ConvLSTM/GNN) nahi. Sirf faasla dekhta hai: deewar ka material, hawa, dhuwan, ya aag ki raftar (waqt) **shamil nahi**. AI wala prediction data ke baghair nahi ban sakta, is liye wo **future work** hai.

Camera ko naqshe se jorna (calibration / homography)

Masla: camera tasveer ko **pixels** mein dekhta hai, naqsha **meters** mein hai. Pehle system ko pata nahi tha ke camera mein dikha insaan naqshe par kahan hai, is liye operator ko haath se click karna padta tha.

Hal: har camera ke liye **4 jodiyan**: farsh ke 4 kone camera ki tasveer mein kahan hain (pixels) aur naqshe par kahan hain (meters). In se ek 3×3 hisaab (**homography**) banta hai jo tasveer ka har farsh wala pixel naqshe ke meter mein badal deta hai. Bilkul jaise mobile ka document scanner tirchi photo ko seedha kar deta hai.

Kaise lagayein (Floor Plans page → Camera calibration):

1. Naqsha kholein aur pehle **Save** karein.
2. **Camera** chunein.
3. Har point ke liye **Camera px** (x, y) likhein. Camera ki ek snapshot Paint mein kholein, kone par mouse rakhein, Paint neeche x, y dikhata hai.
4. **Plan m** likhein, ya **Plan m** ⌖ dabayein aur naqshe par wohi kona click karein.
5. **Save calibration**. Hatana ho to **Remove**.



Is ke baad: dashboard ke Evacuation panel mein us camera ke har insaan ka **amber nuqta** naqshe par aata hai (insaan ke **pairon** ka point: box ka neeche beech). Kisi nuqte par click karein to raasta **usi insaan ki jagah** se nikalta hai.



Cheez	Tafseel
Save kahan	`CameraFeeds.Calibration` column (JSON), migration `CameraCalibration`
API	`PUT /api/cameras/{id}/calibration` (save), `DELETE /api/cameras/{id}/calibration` (hatana)
Jaanch (backend)	Plan maujood ho, 4 points, naqshe ke andar, aur **koi 3 points ek line mein na hon** (warna 400)
Hisaab kahan	Browser mein: `frontend/src/app/shared/utils/homography.ts` (har frame par, server par bojh nahi)
Nuqta kab gayab	Camera se 4.5 second tak frame na aaye

⚠️ **Hadein:** - Sirf **farsh** ke liye sahi hai, is liye pairon ka point. Seedhi ya mezzanine par khade insaan ki jagah ghalat ho sakti hai. - Camera **hila** to calibration dobara karni hogi. - Fish-eye/wide lens mein kinare mude hote hain, wahan andaza kam sahi. - Ek insaan do cameras mein dikhe to **do nuqte**. - Nuqta andaza hai (kuch decimeter tak farq), GPS nahi. - Raasta khud tab nikalta hai jab nuqte par **click** karein; har insaan ka raasta bina click ke nahi. - Calibration save karne par sirf **usi browser** ki list update hoti hai; doosre khule dashboards reload ke baad dekhte hain. - Asli camera ke saath abhi **test nahi** hua: simulator ke nakli insaanon par check kiya.

10. Database (SQL Server: SentinelDb)

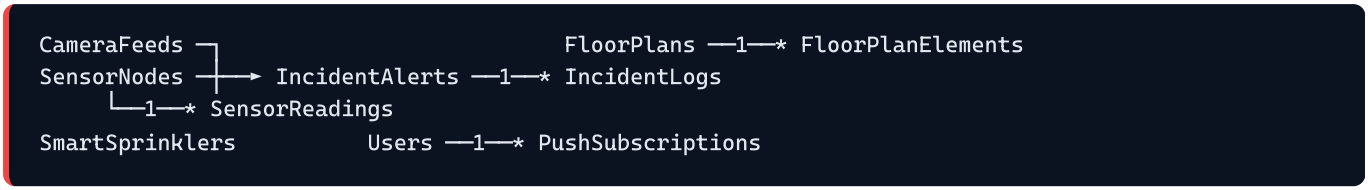


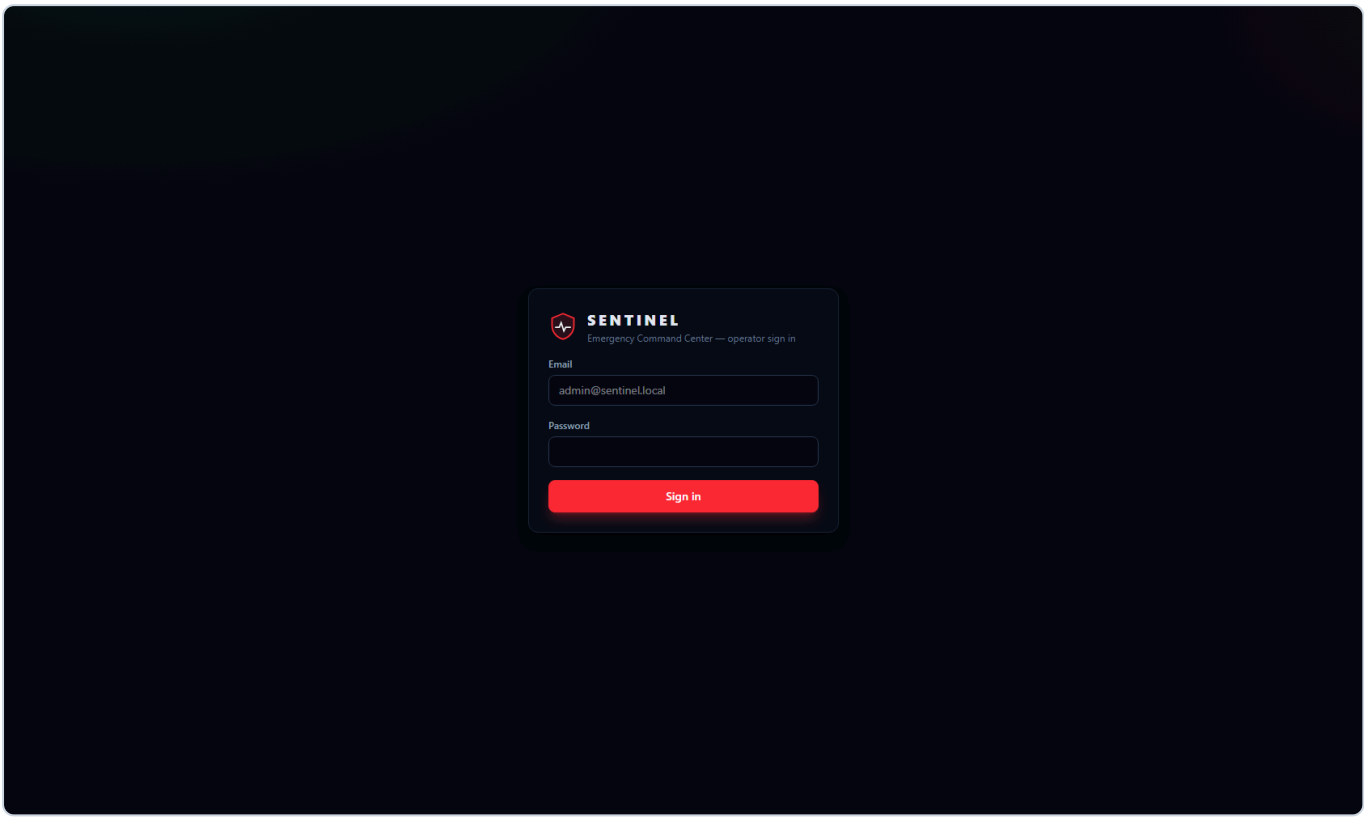
Table	Kya rakhta hai
`CameraFeeds`	Camera: id (`cam-warehouse`), naam, CCTV/Drone, stream URL, zone, naqshe se jor (`Calibration`)
`SensorNodes` / `SensorReadings`	Sensor aur us ki readings ki history
`SmartSprinklers`	Sprinkler: zone, ON/OFF, Auto/Manual
`IncidentAlerts`	Incident: waqt, camera/sensor, zone, severity, aag, insaan, resolve
`IncidentLogs`	Har incident ki history (Chapter 8)
`FloorPlans` / `FloorPlanElements`	Naqshe aur un ke rooms, deewarein, exits, zones
`Users` / `PushSubscriptions`	Login users aur browser push

- Teen migrations: `InitialCreate` , `FloorPlansAndIncidentLogs` , `CameraCalibration` (cameras ka naqshe se jor).
- Waqt **UTC** mein save, browser local time dikhata hai.
- Demo data (Development): 4 cameras (`cam-lobby` Zone A, `cam-warehouse` Zone B, `cam-server` Zone C, `drone-01` Zone B), 6 sensors, 5 sprinklers.

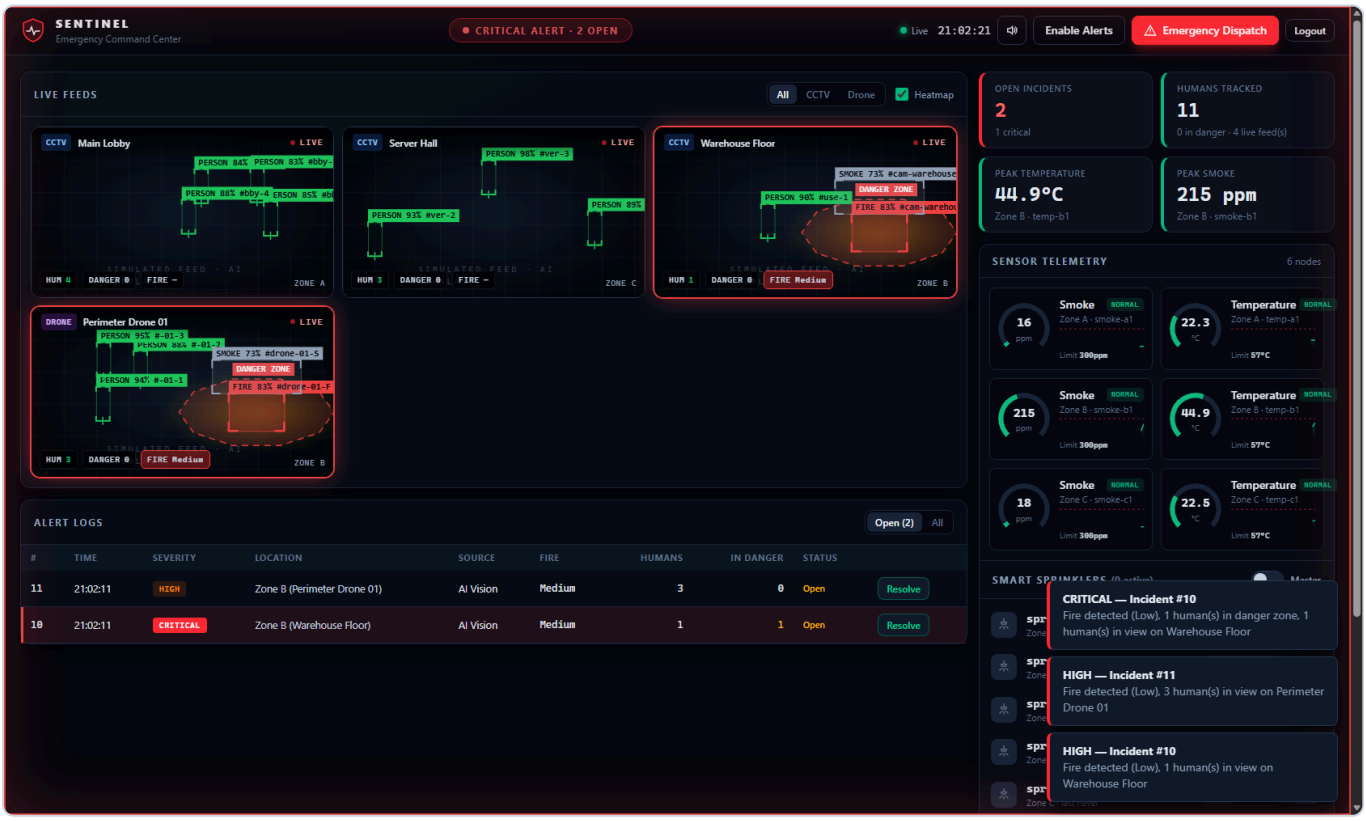
11. Hifazat (Security)

Cheez	Kaise
Operator login	JWT token (8 ghante), password hash
Machines	`X-Device-Key` header
Roles	Operator (dashboard), Device (sensors/AI), OperatorOrDevice
Secrets	`.env` mein, `.git/info/exclude` se git se bahar
SignalR	60 messages/second ki had
Uploads	Sirf PNG/JPG, bytes check, 5 MB, server ka apna file naam
AI service	Sirf `127.0.0.1` par, bahar se nahi jur sakta
AI model files	sha256 check. File badli ho to load nahi hoti
Training checkpoints	`weights_only=True` : sirf numbers khulte hain, koi code nahi chalta
Errors	Har ghalti saaf JSON (400/401/404/500)

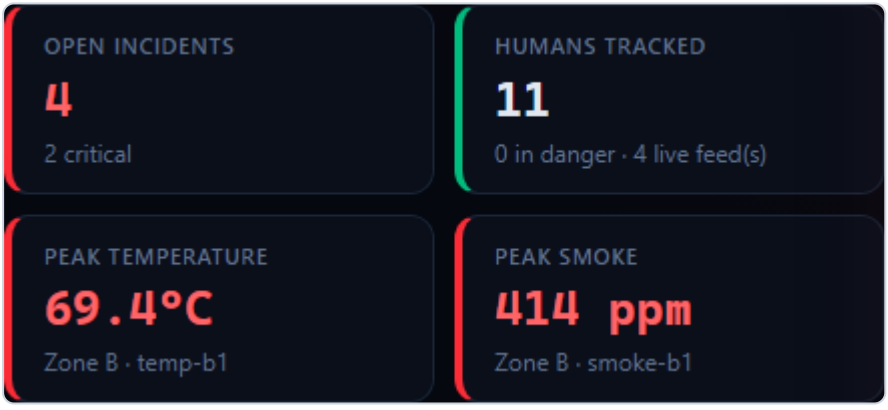
12. Dashboard (Website) Ki Screens



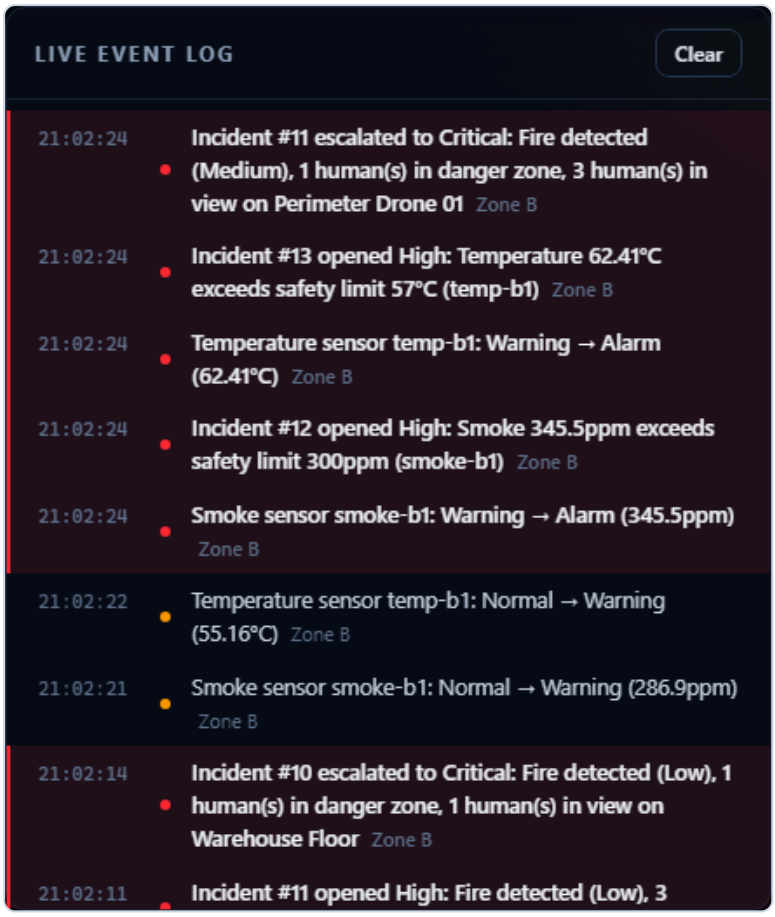
Top bar: system ka haal, Live/Offline, ghari, awaaz band, **Enable Alerts** (browser push), **Emergency Dispatch**, Logout. Menu: **Dashboard · Floor Plans · AI**.

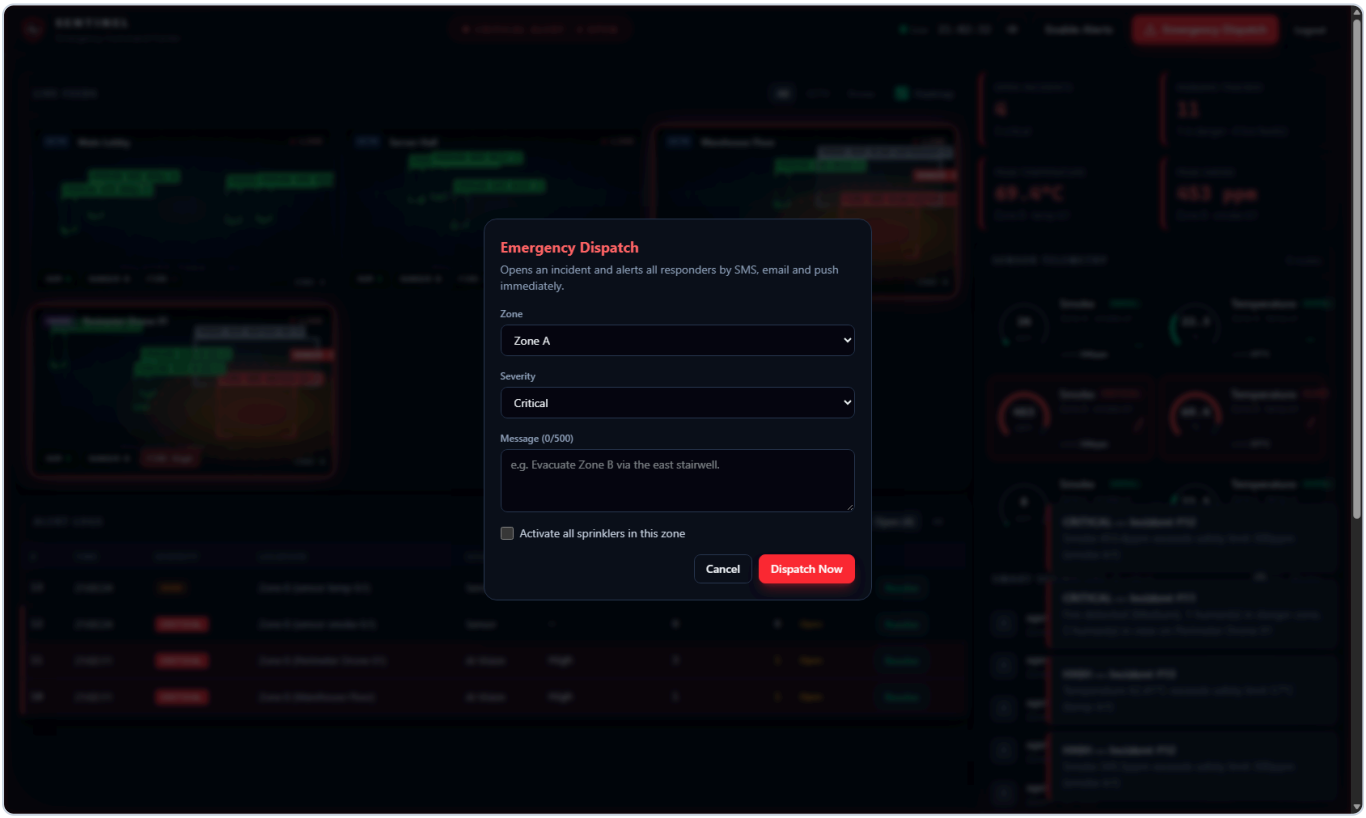


Hissa	Kaam
KPI strip	Khule incidents, ginay gaye insaan, sab se zyada garmi aur dhuwan
Video Wall	Saare cameras, CCTV/Drone filter, heatmap on/off, click = bara
Video tile	Live video + AI boxes, zones, heatmap. Chips: SIM DATA, NO AI DATA
Sensor panel	Gauges, graph, sprinkler switches, Auto/Manual, Master
Alert logs	Khule incidents, Resolve button
Live Event Log	SMS gaya, sprinkler chala, waghera
Incident History	Graph + logs, filters
Evacuation panel	Naqsha, click karke raasta aur ETA, safe zones ki list, calibrated cameras ke insaan (amber nuqte), **fire risk heatmap aur "Next at risk"**
Floor Plans page	Naqsha editor (Chapter 9)
AI page	Models ka haal + test photo (Chapter 20)






LIVE EVENT LOG
Clear
21:02:24 Incident #11 escalated to Critical: Fire detected (Medium), 1 human(s) in danger zone, 3 human(s) in view on Perimeter Drone 01 Zone B
21:02:24 Incident #13 opened High: Temperature 62.41°C exceeds safety limit 57°C (temp-b1) Zone B
21:02:24 Temperature sensor temp-b1: Warning → Alarm (62.41°C) Zone B
21:02:24 Incident #12 opened High: Smoke 345.5ppm exceeds safety limit 300ppm (smoke-b1) Zone B
21:02:24 Smoke sensor smoke-b1: Warning → Alarm (345.5ppm) Zone B
21:02:22 Temperature sensor temp-b1: Normal → Warning (55.16°C) Zone B
21:02:21 Smoke sensor smoke-b1: Normal → Warning (286.9ppm) Zone B
21:02:14 Incident #10 escalated to Critical: Fire detected (Low), 1 human(s) in danger zone, 1 human(s) in view on Warehouse Floor Zone B
21:02:11 Incident #11 opened High: Fire detected (Low), 3


Emergency Dispatch
Opens an incident and alerts all responders by SMS, email and push immediately.
Zone
Zone A
Severity
Critical
Message (0/500)
e.g. Evacuate Zone B via the east stairwell.
Activate all sprinklers in this zone
Cancel
Dispatch Now

13. AI Service: Kya Hai Aur Kaise Kaam Karta Hai

Kya hai? `ai-service/` ek **alag Python program** hai jo camera ki tasveer par AI models chalata hai aur nateeja API ko bhejta hai.

Alag kyun? CPU par ek tasveer 150–450 ms leti hai. API ke andar hota to API slow ho jati. Alag hone se API tez rehti hai, aur AI band ho to baaki system chalta rehta hai.

Cheez	Value
Zubaan	Python 3.13
Libraries	`numpy 2.5.4`, `opencv-python-headless 5.0.0.93`, `onnxruntime 1.31.0` (bas yehi teen)
Hardware	**Sirf CPU.** Purane Intel Xeon W3520 (bina AVX) par chal kar check kiya. ~0.5 GB RAM
Port	`127.0.0.1:8008`
Endpoints	`GET /health` (models ka haal), `POST /infer?cameraId=…` (ek tasveer, max 8 MB)

Har frame par kya hota hai (pipeline)

#	Qadam	Tafseel
1	Letterbox	Tasveer bina khinche 416×416 mein fit
2	Models	Person model + fire/smoke model
3	Filter + NMS	Threshold se kam hatao. Ek cheez par kai boxes hon to sabse acha
4	Tracking	Pichle frame se 30%+ overlap = wohi track id. 5 se zyada frames gayab = track khatam
5	Confirmation	Aag/dhuwan **3 frames** mein dikhe tab report. Insaan foran
6	Severity	Aag ka frame mein area %: 2% se kam Low, 2–10% Medium, 10–25% High, 25%+ Critical
7	AI danger zone	Aag ka box har taraf 25% bara
8	Heatmap	16×9 khane, har frame 0.8 se thanda. **Sirf "kahan dikha"**, prediction nahi
9	Explanation	Misaal: `fire 0.83 ≥ threshold 0.40; box 3.1% of frame; confirmed: track F11 seen 3/4 frames`
10	Bhejna	`POST /api/detections` + `X-Device-Key`

Simulated: video **file** se aaya data hamesha `simulated: true`. Live camera (RTSP) par aap khud batate hain.

14. AI Model 1: Person (Insaan) Model

Sawal	Jawab
Model	**YOLOX-Tiny** (object detection)
Kis ne banaya	**Megvii** (company) ki research team, open-source project "YOLOX"
Kahan se liya	Official GitHub release: `github.com/Megvii-BaseDetection/YOLOX/releases/tag/0.1.1rc0`, file `yolox_tiny.onnx`
Kis par trained	**COCO dataset** (Microsoft COCO, ~118,000 training images, 80 qisam ki cheezein)
Humne kya kiya	Khud train **nahi** kiya. Official model jaisa hai waisa liya. 80 classes mein se sirf `person` istemaal
License	Apache-2.0 (free, commercial bhi)
Input	416 × 416 pixel
Threshold	0.45
File / size	`ai-service/models/yolox_tiny.onnx`, ~20 MB
sha256	`427cc366d34e27ff7a03e2899b5e3671425c262ea2291f88bb942bc1cc70b0f7`
Repo mein	Haan, `ai-service/models/` mein. File na ho to `tools/download_models.py` official link se laata hai (sha256 check)

Test: Penn-Fudan dataset

Cheez	Value
Dataset	**Penn-Fudan Pedestrian Database** (University of Pennsylvania), `cis.upenn.edu/~jshi/ped_html/`
Size	170 tasveerein, hamare evaluation mein 423 insaan ke boxes
Kami	Har tasveer mein insaan hain, is liye "khali scene" par galti nahi napi gayi

Threshold	Precision	Recall	F1
0.25	0.618	0.993	0.762
0.35	0.663	0.993	0.795
0.45 (laga hua)	0.707	0.993	0.826
0.55	0.758	0.991	0.859

- **Recall 0.993:** 423 mein se 420 insaan pakde ✅ (shart ≥ 0.85).
- **Speed:** p95 139 ms ✅.
- **Precision 0.707** (shart ≥ 0.80): "galat" boxes aankh se dekhe gaye. Zyada tar **asli insaan** the (cycle wale, peeche khade) jin par Penn-Fudan ne label nahi lagaya. Phir bhi 0.80 **pakka nahi**.

- **Faisla:** *improvement required*. Agla qadam: poore label wala dataset (COCO val2017 ya site CCTV).

15. AI Model 2: Fire/Smoke (Aag/Dhuwan) Model: Kya Aur Kahan Se

Khud train kyun karna para?

- Person model (COCO) mein aag ya dhuwen ki class **hai hi nahi**.
- Koi official, free license wala, tayyar fire/smoke ONNX model nahi mila.
- D-Fire walon ke apne YOLOv5 weights hain, lekin wo `.pt` pickle hain (khulte waqt code chal sakta hai) aur YOLOv5 ka license AGPL-3.0 hai. Is liye nahi liye.

Model ki pehchaan

Sawal	Jawab
Model	**YOLOX-Tiny** (wohi design jo person model ka), depth 0.33, width 0.375
Shuru ka model (base)	Official **COCO pretrained** `yolox_tiny.pth`, Megvii release 0.1.1rc0
Base ka sha256	`9de513de589ac98bb92d3bca53b5af7b9acfa9b0bacb831f7999d0f7afaee8f0`
Tareeqa	**Fine-tuning:** COCO par seekha hua model aag/dhuwan ke liye dobara sikhaya
Training code	YOLOX 0.3.0 (`github.com/Megvii-BaseDetection/YOLOX`), Apache-2.0
Classes	2: **0 = smoke (dhuwan)**, **1 = fire (aag)**
Input	416 × 416 pixel
Threshold	0.40
Natija file	`ai-service/models/fire_smoke_yolox.onnx`, ~20 MB
sha256	`8d73df91dbb0241cfb04cacf529a8be4b26e68d4fccbb20cd816884bfa914d73`
License	YOLOX code Apache-2.0 + D-Fire data CC0

16. Dataset: D-Fire (Kahan Se, Kitni Images)

Sawal	Jawab
Naam	**D-Fire**: An image dataset for fire and smoke detection
Kahan se	`github.com/gaiasd/DFireDataset` (official OneDrive link)
Banane wale	Pedro de Venâncio, Adriano Lisboa, Adriano Barbosa (Brazil). Paper: *Neural Computing and Applications*, 2022
License	**CC0 1.0** (public domain, bilkul free)
File	`D-Fire.zip`, 3,036,222,313 bytes (~3 GB)
sha256	`3824fb3ce32cfa8b538792dfd460603648d271072ac6ae34f1af0c713f60260c`
Label format	YOLO: har tasveer ki `.txt` file, har line = `class cx cy w h`
Kul tasveerein	**21,527** (train 17,221 + test 4,306)

Tasveeron ki ginti (humne khud gini)

Hissa	Tasveerein	Bina aag/dhuwen	Aag wali	Dhuwen wali	Aag ke boxes	Dhuwen ke boxes
Official TRAIN (poora)	17,221	7,833	4,707	8,444	11,814	9,550
→ **Training** (85%)	**14,621**	7,123	3,696	6,755	9,125	7,577
→ **Validation** (15%)	**2,600**	710	1,011	1,689	2,689	1,973
Official **TEST**	**4,306**	2,005	1,115	2,081	2,878	2,315

(Ek tasveer mein aag aur dhuwan dono ho sakte hain, is liye columns ka jama tasveeron se zyada hai.)

Data ki jaanch aur bantwara (`tools/prepare_dataset.py`)

- Har tasveer khol kar dekhi, labels ki range check ki.
- Train aur test mein **ek bhi same tasveer nahi** (leakage nahi).
- 26 khali (zero-area) labels hata kar gine gaye.
- **Validation** official TRAIN mein se nikali: file naam ke **groups** (100 lagatar frames ka ek group, kul 174 groups → 148 training, 26 validation), seed 42. Is se ek hi jagah ki milti julti tasveerein dono taraf nahi jaatin.
- **TEST set training mein kabhi use nahi hua.** Sirf aakhri score ke liye.

17. Training: Kahan, Kaise, Kitne Epochs

Pehli koshish: is computer par CPU pilot (fail)

Cheez	Value
Hardware	Intel Xeon W3520 (bina AVX), GPU kaam ka nahi (Quadro FX 380, 256 MB)
Andaza	YOLOX-Tiny 80 epochs = **~13.6 din** (0.867 s per image)
Pilot	YOLOX-Nano @ 320, 4,000 training + 800 validation tasveerein, 4 epochs (lr 0.01, backbone frozen)
Natija	Aag recall **0.000**: kuch nahi pakda. **Install nahi** kiya

Asal training: Kaggle GPU par

Cheez	Value
Jagah	**Kaggle** (Google ki free cloud), notebook "Sentinel", Version 1 ("Save & Run All")
GPU	**NVIDIA Tesla T4**
Tareekh	14 October 2026
Notebook	`ai-service/training/train_fire_smoke_colab.ipynb`
Settings file	`ai-service/training/yolox_tiny_dfire.py`
Training tasveerein	**14,621**
Validation tasveerein	**2,600**
Epochs	**80**
Batch size	32 tasveerein ek saath
Steps per epoch	457 (14,621 ÷ 32)
Kul steps	36,560 (457 × 80)
Speed	~0.45 second per step, ~3.5 minute per epoch
Kul waqt	**~4 ghante 39 minute**
Precision	fp16 (mixed precision, tez)
Augmentation	Mosaic (scale 0.5–1.5), random size (320–640), aakhri **10 epochs bina augmentation**, mixup band
Validation	Har 5 epoch baad. Best = sab se acha validation AP
Loss	13.3 (shuru) → ~5.5 (epoch 2 tak)
Best validation	**AP50:95 = 0.359**, **epoch 80** par
Best checkpoint	`best_ckpt.pth`, 40.58 MB, sha256 `df57126a47872a8e3899486d1d2ff45962d28a7257363a522dcc1e9c99d32ffc`

Ek masla aur us ka hal

- Training **poori** hui, lekin notebook ka "export" cell fail hua: YOLOX 0.3.0 ka export `torch.onnx._export` use karta hai jo naye PyTorch mein nahi.
- Checkpoints Kaggle Output mein mehfooz the. `best_ckpt.pth` download kiya.
- Apna script `ai-service/training/export_onnx.py` banaya:
  - `torch.load(weights_only=True)`: sirf numbers khulte hain, koi code nahi chalta. Sirf numpy ka number type allow (YOLOX `best_ap` usi mein rakhta hai).
  - `torch.onnx.export` (opset 11) → `fire_smoke_yolox.onnx` → sha256 `models.json` mein.

```
cd ai-service\training
python export_onnx.py --exp yolox_tiny_dfire.py --ckpt best_ckpt.pth --out
..\models\fire_smoke_yolox.onnx
```

18. Accuracy: Asli Numbers

Lafzon ka matlab

Lafz	Matlab
Recall	Jitni aag thi us mein se kitni pakdi. 0.93 = 100 mein se 93
Precision	Model ke boxes mein se kitne sahi
FPR (galat alarm)	Bina aag wali tasveeron mein se kitni par galti se aag
Image-level	Poori tasveer: aag hai ya nahi
Box-level	Har shola/dhuwan sahi jagah par (IoU ≥ 0.5)
AP50:95	Training ka mila jula score (box kitna theek baitha)

Pass hone ki shartein

Class	Shart
Fire	Image recall ≥ 0.90 **aur** FPR ≤ 0.10
Smoke	Image recall ≥ 0.80, FPR ≤ 0.15
Fire/Smoke boxes	Precision ≥ 0.60
Person	Recall ≥ 0.85 aur precision ≥ 0.80
Speed	p95 ≤ 1000 ms

Fire/Smoke: D-Fire TEST (4,306 tasveerein, training mein kabhi nahi dekhi)

Threshold	Fire box P / R	Fire image recall / FPR	Smoke box P / R	Smoke image recall / FPR
0.25	0.642 / 0.697	0.952 / 0.011	0.793 / 0.798	0.947 / 0.039
0.30	0.676 / 0.682	0.947 / 0.009	0.808 / 0.791	0.939 / 0.034
0.35	0.706 / 0.668	0.940 / 0.007	0.821 / 0.784	0.932 / 0.032
0.40 (laga hua)	0.731 / 0.649	**0.932 / 0.006**	0.833 / 0.777	**0.923 / 0.028**
0.50	0.783 / 0.602	0.904 / 0.004	0.856 / 0.754	0.891 / 0.024

Shart (threshold 0.40)	Mila	
Fire image recall ≥ 0.90	**0.932** (1,115 mein se 1,039)	✅
Fire galat alarm ≤ 0.10	**0.006** (2,005 mein se 4)	✅
Smoke image recall ≥ 0.80	**0.923** (2,081 mein se 1,920)	✅
Smoke galat alarm ≤ 0.15	**0.028** (2,005 mein se 17)	✅

Shart (threshold 0.40)	Mila	
Box precision ≥ 0.60	Fire 0.731, Smoke 0.833	✅
Speed p95 ≤ 1000 ms	**356 ms** (CPU)	✅

Mushkil cases (0.40):

- Sirf dhuwan (aag nahi): 1,186 mein se 1,074 par dhuwan pakda, 15 par galti se aag.
- Chhoti aag (frame ka 1% se kam): 831 mein se 769.
- Andheri tasveerein: 229 bina aag wali mein se 2 par galat aag.

Speed (Xeon CPU): fire model akela mean 182 ms, p95 356 ms. Dono models saath ~460 ms per frame.

CPU pilot se muqabla: aag recall 0.000 → **0.932**.

Faisla: *meets the defined development criteria* (D-Fire test par).

> ⚠️ **Kamzoriyan:** - **Fire box recall 0.65:** tasveer mein aag pakad leta hai, lekin har shola nahi. Severity kabhi kam aa sakti hai. - D-Fire test usi qisam ki tasveeron se hai jaisa train. Site ka CCTV, fog, steam, dust, IR **test nahi**. - Ye lab ke numbers hain, **safety certificate nahi**.

19. End-to-End Test: Poora System Ek Saath

Maqsad: model akela nahi, **poora system** check karna: AI service → API → database → SignalR → dashboard.

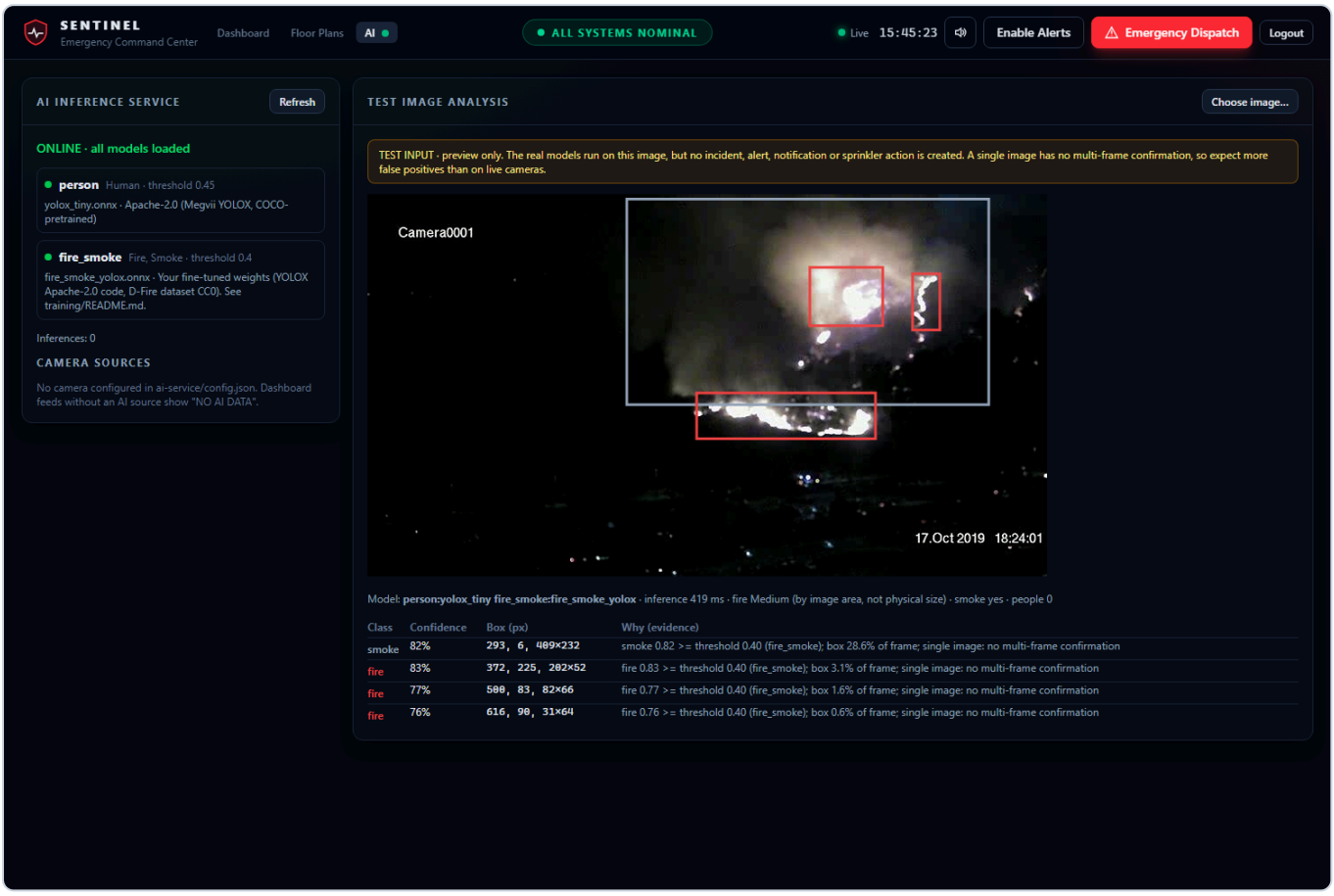
Hifazat:

- Alag test database `SentinelDb_E2E_20261015`, baad mein **delete**. Asli `SentinelDb` ko nahi chhua.
- `.env` **load nahi** ki (asli SMS/Email credentials), throwaway keys. Koi hardware nahi.
- Input `e2e-fire.avi`: D-Fire TEST ki 40 tasveerein ki video, camera `cam-warehouse` (Zone B), **SIMULATED**.

Run	Kya hua
Run 1 (1 frame/second)	10 frames, **0 detection**. Detector ne har frame mein aag pakdi, lekin ye alag alag photos hain, aag ki jagah har baar badli, to 3-frame confirmation ne sahi taur par kuch report nahi kiya. **Sabaq:** camera ka view 3 frames tak ek jaisa hona chahiye
Run 2 (4 frame/second)	40 frames, 0 failure. Incident #1: **Low** → **Medium**, "Fire detected (Medium), Smoke detected … [SIMULATED DATA]". **Sirf ek** incident (dedupe). 6 logs (fire 1.05% → 2.5% → 4.36% → 0.54%), sab simulated. SignalR events dashboard tak. **Paanchon sprinkler band**

Test nahi hua: emergency raasta (High/Critical, danger zone mein insaan) jo notifications aur sprinklers chalata hai. Wo sirf unit tests mein.

20. AI Page (Dashboard Par)



Baen taraf, "AI inference service":

- Hara **ONLINE · all models loaded** / Peela *degraded* (koi model missing) / Laal **OFFLINE**.
- Har model: naam, classes, threshold, file, license. Kitni inferences, average ms.
- Camera sources: kaun se camera jure, haal, frames. File par **SIMULATED / TEST INPUT**.

Daen taraf, "Test image analysis":

1. **Choose image...** (PNG/JPG, 5 MB tak).
2. Asli models chalte hain, boxes tasveer par (aag laal, dhuwan gray).
3. Table: class, confidence, box, **"Why (evidence)"**.

> ⚠️ Sirf **preview**: koi incident, alert ya sprinkler nahi. Kuch na mile to iska matlab ye **nahi** ke jagah mehfooz hai.

APIs: `GET /api/ai/status`, `POST /api/ai/analyze` (multipart `file`). Errors: 503 service/model nahi, 504 timeout, 502 kharab jawab.

21. Chalane Ka Tareeqa (Setup)

Pehle install: .NET 8 SDK, Node.js 20.19+ aur Angular CLI, SQL Server, `dotnet-ef`, Python 3.13.

`.env` (project folder mein, `.env.example` se copy)

Key	Kaam
`ConnectionStrings__Sentinel`	SQL Server
`Jwt__SigningKey`	Login token ki chaabi (lambi random)
`Device__ApiKey`	Machines ki chaabi
`Seed__AdminEmail`, `Seed__AdminPassword`	Pehla admin
`Notifications__Sms__…`, `__Email__…`, `__WebPush__…`	Alerts (ikhtiyari)

4 windows

```
# 1) Backend API  → http://localhost:5291/swagger
cd <project>\backend
dotnet ef database update --project Sentinel.Api
dotnet run --project Sentinel.Api --launch-profile http

# 2) Dashboard  → http://localhost:4200
cd <project>\frontend
npm install
ng serve

# 3) AI service  → http://127.0.0.1:8008/health
cd <project>\ai-service
py -3.13 -m venv .venv
.\.venv\Scripts\python -m pip install -r requirements.txt
.\.venv\Scripts\python -m sentinel_ai          # dono models repo mein hain, download ki zaroorat
nahi

# 4) (Demo) Simulator: Zone B mein aag
cd <project>\backend
dotnet run --project Sentinel.Simulator -- --device-key <Device__ApiKey> --ignite "Zone B"
```

Simulator ke options: `--base-url`, `--device-key`, `--ignite`, `--fps`, `--fire-probability`, `--telemetry-interval`.

AI se camera jorna (`ai-service/config.json`)

```
{
  "api": "http://localhost:5291",
  "cameras": [
    { "cameraId": "cam-lobby", "source": "${CAM_LOBBY_URL}", "fps": 1, "enabled": true }
  ]
}
```

- `source`: `rtsp://…` (live), `0` (webcam) ya video file (hamesha SIMULATED).
- Camera ka password environment variable mein, `SENTINEL_DEVICE_KEY` = `Device__ApiKey`.
- Dono models (`yolox_tiny.onnx`, `fire_smoke_yolox.onnx`) repo ke `ai-service/models/` mein hain.

> ⚠️ Asli camera aur asli `.env` ke saath aag dikhi to **asli SMS/Email** jayenge. Pehle safe test number/email se test karein.

Database doosre PC par (office ↔ ghar)

Database git ke zariye ek backup file `database/SentinelDb.bak` ban kar jata hai.

```
# Kaam khatam (PC 1)
.\database\save.ps1
git add database\SentinelDb.bak; git commit -m "db: update"; git push

# Kaam shuru (PC 2) — API band karke; ye local database ko POORA badal deta hai
git pull
.\database\load.ps1          # "yes" likhein
```

Baat	Matlab
Database nahi hai	`load.ps1` naya bana deta hai
Database hai	Backup se **poora badal** deta hai (mila kar nahi)
Aakhri `save` jeetta hai	Ek waqt mein ek hi PC par kaam karein
SQL Server	Doosre PC par 2022 ya naya; Express ho to `-Server "localhost\SQLEXPRESS"`
Login	Users bhi saath aate hain: wohi email/password chalta hai
`.env`	Har PC par alag banti hai, git mein nahi jaati

22. Tests

Hissa	Tests	Kya check
Backend (xUnit)	**60**	Fire risk prediction, alert rules, API + SignalR integration, evacuation router, floor plans + logs, camera calibration, camera lost, simulated sensors, AI client
Frontend (Vitest)	**26**	Fire risk list, login expiry, homography, calibrated camera ke nuqte, overlay drawing, floor plan editor, evacuation panel, AI page, formatting
AI service (unittest)	**15**	Box decode, NMS, kharab/badli model file rad, tracking, 3-frame confirmation, severity, adhoori settings, kharab frame ke baad camera chalta rahe, API contract, service errors, video = SIMULATED (mock: sirf logic, accuracy nahi)
Production build	✅	`ng build`
AI evaluation	Asli models	Chapter 14, 18
End-to-end	Asli services	Chapter 19

```
cd backend;    dotnet test
cd frontend;   ng test --watch=false;  ng build
cd ai-service; .\.venv\Scripts\python -m unittest discover -s tests
cd ai-service; .\.venv\Scripts\python tools\evaluate.py --data <D-Fire\test> --classes
0=Smoke,1=Fire --sweep 0.25,0.3,0.35,0.4,0.5 --out reports\fire
```

Sab 15 October 2026 ko pass hue.

23. API Aur SignalR Ki Poori List

Method	Route	Kaun	Kaam
POST	`/api/auth/login`	Sab	Login → JWT
GET	`/api/auth/me`	Login	Apni maloomat
GET	`/api/dashboard/snapshot`	Operator	Poori screen ka data
GET	`/api/cameras`, `/api/cameras/{id}`, `/api/cameras/{id}/stream`	Operator/ Device	Cameras, MJPEG stream
POST	`/api/detections`	Operator/ Device	AI frame
GET	`/api/incidents`, `/api/incidents/{id}`	Operator	Incidents
GET	`/api/incidents/logs?incidentId=&type=&limit=`	Operator	Logs
POST	`/api/incidents/{id}/resolve`	Operator	Band karna
POST	`/api/incidents/dispatch`	Operator	Manual emergency
GET	`/api/iot/sensors`, `/sensors/{id}`, `/sensors/{id}/history`	Operator/ Device	Sensors
POST	`/api/iot/sensors/{id}/readings`, `/api/iot/telemetry` (`simulated: true` = test data)	Operator/ Device	Readings
GET	`/api/iot/sprinklers`	Operator/ Device	Sprinklers
POST	`/api/iot/sprinklers/{id}/toggle`, `/api/iot/sprinklers/master`	Operator/ Device	ON/OFF
GET/POST/PUT/ DELETE	`/api/floorplans`, `/api/floorplans/{id}`	Operator	Naqshe
PUT/GET	`/api/floorplans/{id}/image`	Operator	Naqshe ki tasveer
POST	`/api/floorplans/{id}/route`	Operator	Evacuation raasta
PUT/DELETE	`/api/cameras/{id}/calibration`	Operator	Camera ko naqshe se jorna / hatana
GET	`/api/floorplans/{id}/risk`	Operator	Aag kin kamron tak pahunch sakti hai + risk heatmap
GET	`/api/ai/status`	Operator	AI ka haal
POST	`/api/ai/analyze`	Operator	Test photo
GET/POST	`/api/push/vapid-public-key`, `/subscribe`, `/unsubscribe`	Operator	Browser push

Method	Route	Kaun	Kaam
GET	`/health`	Sab	API zinda hai?

SignalR hub `/hubs/monitoring`:

- **Server → browser events:** `snapshot`, `detection`, `telemetry`, `sprinkler`, `incident`, `notification`, `activity`, `incidentLog`.
- **Device → server methods:** `PublishDetection`, `PublishTelemetry`.

24. Ahem Files

File	Kaam
`backend/Sentinel.Api/Program.cs`	App ka switchboard: `.env`, DI, auth, CORS, SignalR, Swagger
`Services/DetectionService.cs`	Detection: validate → broadcast → evaluate
`Services/AlertEngine.cs`	Severity, incidents, emergency, logs
`Services/SensorService.cs`, `SensorRules.cs`	Sensor readings aur halat
`Services/SprinklerService.cs`	Sprinklers
`Services/Notifications/`	SMS, Email, Push
`Services/FloorPlanService.cs`, `EvacuationRouter.cs`	Naqshe aur raasta
`Services/AiInferenceClient.cs`, `Controllers/AiController.cs`	API ↔ AI service
`Validators/DetectionPayloadValidator.cs`	AI JSON jaanch + danger zone person check
`Hubs/EmergencyMonitoringHub.cs`	SignalR
`frontend/src/app/core/services/signalr.service.ts`	Live data
`frontend/src/app/features/dashboard/video-feed/overlay-renderer.ts`	Boxes/zones/heatmap drawing
`frontend/src/app/features/floor-plans/`	Naqsha editor
`frontend/src/app/features/ai/`	AI page
`frontend/src/app/shared/utils/homography.ts`	Camera pixel → naqsha meter (4 points se)
`backend/Sentinel.Api/Services/FireRiskEstimator.cs`	Aag ke phailne ka rule-based andaza + heatmap grid
`ai-service/sentinel_ai/detector.py`	ONNX model chalana + sha256
`ai-service/sentinel_ai/pipeline.py`, `tracker.py`	Confirmation, severity, zones, heatmap, tracking
`ai-service/sentinel_ai/service.py`	HTTP server + camera workers
`ai-service/models/models.json`	Models ki list
`ai-service/tools/prepare_dataset.py`, `evaluate.py`	Dataset jaanch, accuracy
`ai-service/training/`	Notebook, settings, export script
`ai-service/reports/`	Evaluation reports (JSON)

25. Requirements Ka Status

Kul 26 points: ✅ 22 · ⚠️ 4 · ❌ 0

#	Requirement	Status	Note
1.1	Video se aag/dhuwan	✅	Fire/smoke model, 93% / 92%
1.2	Boxes/heatmap dikhana	✅	
1.3	Mukhtalif mahol	⚠️	Sirf andheri photos. Fog, steam, IR nahi
1.4	Accuracy	✅	Chapter 18
2.1	Insaan pehchaanna	⚠️	Recall 99%, precision 0.80 pakki nahi
2.2	Insaan track karna	⚠️	IoU tracker, bheer mein IDs badal sakti hain
2.3	Danger zone mein insaan	✅	Backend ka polygon check
2.4	Evacuation raasta	✅	Dijkstra, 1 m clearance
2.5	Khatre saaf nishaan-zada	✅	
3.1	Aag small/medium/large	✅	AI area % se
3.2	Aag ka barhna	✅	Graph, physical size nahi
3.3	Phailne ka prediction	✅	Rule-based: darwazon ke raaste faasla se agle kamre. AI (ConvLSTM/GNN) future work
3.4	Risk heatmap	✅	Naqshe par risk heatmap (laal → narangi → peela), camera par "kahan dikha" heatmap bhi
3.5	Early warning	✅	
4.1–4.4	Alerts	✅	Asli bhej kar test nahi
5.1	Live video par nishaan	✅	Asli camera test nahi
5.2	Safe zones	✅	
5.3	Logs	✅	
5.4	Dashboard	✅	
6.1	Drone feeds	✅	
6.2	IoT/sprinkler simulation	✅	
6.3	Explainable AI	⚠️	Likhi hui wajah, visual nahi
6.4	Evacuation time	✅	Andaza

26. Kya Reh Gaya (Functionality)

#	Functionality	Abhi	Hona chahiye
1	**Camera lost alert**	✅ Ban gaya: 30 sec chup = alert, zone mein aag ho to High	Asli camera ki bijli kaat kar test
2	**Asli CCTV camera**	Sirf video file/photos	RTSP camera live
3	**Asli alerts pahunchna**	Code bana, bhej kar check nahi	Test number/email par tasdeeq
4	**Aag phailne ka prediction**	✅ Rule-based ban gaya (faasla + darwaze)	AI model (ConvLSTM/GNN) jab data ho
5	**Camera ↔ naqsha**	✅ Ban gaya: calibrated camera ka insaan naqshe par nuqta, click = raasta (asli camera par test baqi)	Har insaan ka raasta bina click ke; aag ki jagah bhi naqshe par
15	**Simulated sensor data**	✅ Ban gaya: simulator ki readings SIMULATED, sprinklers khud nahi chalte	—
6	**Tircha kamre + seedhiyan**	Seedhe kamre, har floor alag	Har shakal, floors ke beech raasta
7	**Smart ETA**	Faasla ÷ 1.2 m/s	Bheer, dhuwan, seedhiyan
8	**Raat/fog/steam AI**	Sirf andheri photos check	IR/fog footage par test aur training
9	**Insaan AI ki jaanch**	Precision pakki nahi	Poore label wale data par test
10	**Bheer mein tracking**	Saada tracker	ByteTrack jaisa
11	**Visual explanation**	Likhi wajah	Grad-CAM
12	**Severity calibration**	Screen ka %	Har camera ke liye meters
13	**Rescue ko raasta SMS**	Nahi	Raasta SMS mein
14	**Kai servers**	Dedupe ek process mein	Shared lock (Redis)

Aakhri baat

- AI **madadgar** hai, **fire alarm ka badal nahi**.
- Aag mein bijli jaye to camera band ho sakte hain: cameras UPS/PoE backup par.
- Asli fire-safety hardware ko test mein bina ijazat aur safe tareeqe ke na chalayein.
- Site par test aur certified fire-safety review ke baghair system **production-safe nahi**.